CSC 3320: System-Level Programming

Chapter 2: UNIX Utilities for Non-programmers – Part II

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Note: These slides are updated based on original notes from Raj Sunderraman, Michael Weeks, and Yuan Long

Announcements

Labs:

- For lab related inquiries, always check with Lab TAs if you have any confusion. Keep me in cc.

Homework 1 will be out by Today, Deadline: January 30, 11:59 PM

The vi Editor

visual text editor

- Very common on Unix systems
- Created by Bill Joy in 1976
- Many versions exist, including vim
  - vi improved

The vi Editor

Command mode vs Text Entry Mode

- To enter text entry mode use the following commands:
  - i,I,a,A,o,O,R
- to get back to command mode use ESC

line ranges: 1,$ 1,. .,$.,.-2 5,12 .,.+12

http://www.lagmonster.org/docs/vi.html

Examples

- Example: :5 moves to line 5.
- Example: :3,7d deletes lines from 3 to 7.
- .: Refers to the current line.
- $: Refers to the last line in the file.
- + or -: Moves the range up or down relative to the current line.
- :.,$d deletes from the current line to the end of the file.
- :.,+3d deletes from the current line to three lines below.

The vi Editor

- Example: :1,3t 10 copies lines 1 to 3 and pastes them after line 10. (t basically indicates transfer/copy)
- Example: :5,10m 15 moves lines 5 to 10 after line 15.
- :start,ends/pattern/replacement/flags
  - Example: :1,10s/foo/bar/g
  - replaces "foo" with "bar" in lines 1 to 10.

The vi Editor

Cursor Movement:

up arrow	k
down arrow	j
left arrow	h
right arrow	l
beginning of line	0
first non-whitespace char in line	^
end of line	$

The vi Editor

Cursor Movement:

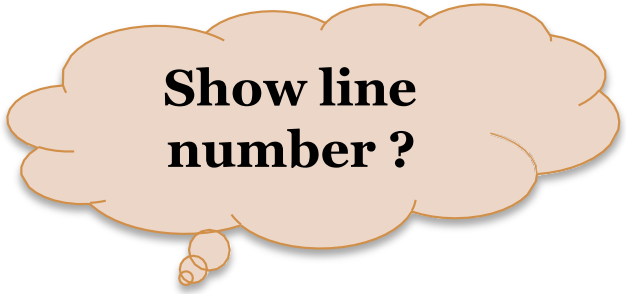
- next word — w
- previous word (back) — b

Goto line number *n*

- :n

Go to last line

- :$



Answer :set number

The vi Editor

CTRL-D (down half screen)

CTRL-U (up half screen)

CTRL-F (forward one screen)

CTRL-B (back one screen)

The vi Editor

Deleting Text:

- delete character under cursor — x
- delete word — dw
- delete line — dd (also D)
- delete many lines — :<range>d

Replacing text:

- replace char under cursor — r
- replace word under cursor — cw
- replace entire line — cc
- substitute next 4 characters — 4s

The vi Editor

Pasting text:

- :<range>y (yank lines)
- 5yy (yank 5 lines)
- p (paste from buffer after cursor)
- P (paste from buffer before cursor)
- **:n** then **p** (after line *n*) (first :n, then press p)

The vi Editor

Searching:

- search forward /w
- search backward ?w
- next n
- next in opposite direction N

Searching/Replacing:

- :<range> s/old/new/ **Next occurrence in current line(s)**
- :<range> s/old/new/g **All occurrences in current line(s)**
- : ~~<range>~~ %s/old/new/g **All occurrences in file**

The vi Editor

Saving/Loading:

write to file	:w <fname>
write again	:w
write	:<range>w <fname>
write and quit	:wq
edit new file	:e <fname>
read (insert) a file	:r <fname>

The vi Editor

Misc

- re-draw screen — CTRL-L
- execute a shell command — :!command

Quitting:

- quit — :q
- write and quit — :wq
- quit without writing — :q!
- exit and write if changed — :x